

TP4

Bio-Informatique

NGS : Contrôle Qualité des Données (FastQC)

Nome et Prénom	Grp	Filière
Belkadi Hamza	G1	IA

Partie 1 :

The screenshot shows the Galaxy Europe interface. On the left is a sidebar with navigation icons. The main panel displays a dataset named 'test' (2.5 MB, format: fastqsanger.gz, database: ?, size: 2.5 MB). Below the dataset information, there is a 'Tool Parameters' section with input parameters like 'request_version' and 'request_json'. On the right, a 'History' panel shows a list of datasets, including 'test' and several 'FastQC' reports.

Dataset Information

Property	Value
Number	1
Name	test
Created	Monday Dec 15th 11:17:13 2025 GMT+1
Filesize	2.5 MB
Dbkey	?
Format	fastqsanger.gz
File contents	contents
History Content API ID	26c75dccc6b16ac8d6c6c10464b45b71
History API ID	11ac94876d9bb33a19fe6e7ec97a2288 (view)
UUID	73f5619b-5c43-4eea-bad9-93a52361cf98
Full Path	/data/dnb12/galaxy_db/files/7/3/1/dataset_73f5619b-5c43-4eea-bad9-93a52361cf98.dat
Sources	https://zenodo.org/record/5189263/files/CV1537_S1_L1_001_R1_001.fastq.gz
Hashes	3abbe7be32ad58e2e0c12cef4ce236961327b6b1d38342e9a00453192312ebd

Tool Parameters

Input Parameter	Value
request_version	1
request_json	[{"targets": [{"destination": "type": "hdsas", "elements": [{"name": "test", "dbkey": "T", "text": "auto", "space_to_tab": false, "to_posix_lines": true, "deferred": false, "src": "url", "url": "https://zenodo.org/record/5189263/files/CV1537_S1_L1_001_R1_001.fastq.gz", "hashes": []}], "type": "hdsas"}], "type": "hdsas"}]

Partie 2 :

The screenshot shows the Galaxy Europe interface with a FastQC report for the dataset 'test'. The report is titled 'FastQC Report' and includes a 'Summary' section with a list of checks. The 'Basic Statistics' table shows file information. The 'Per base sequence quality' section includes a bar chart showing quality scores across all bases.

FastQC Report Summary

- Basic Statistics
- Per base sequence quality
- Per sequence quality scores
- Per base sequence content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Adapter Content

Basic Statistics

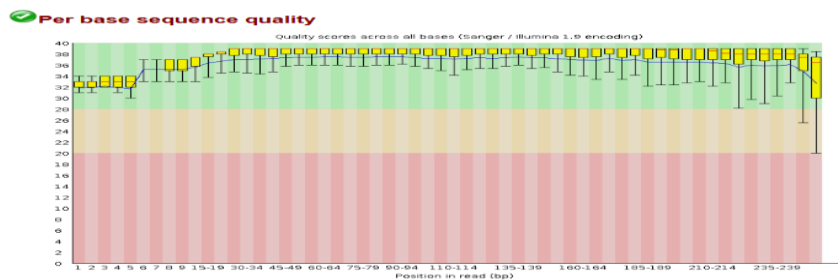
Measure	Value
Filename	test.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	17216
Total Sizes	3.8 Mb
Sequences flagged as poor quality	0
Sequence length	25-251
N/C	38

Per base sequence quality

Quality scores across all bases (Sanger / Illumina 1.9 encoding)

Partie 3 :

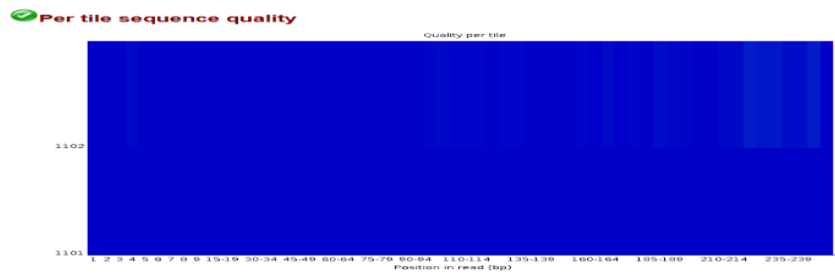
Qualité par base



- Oui - les scores sont élevés et constants

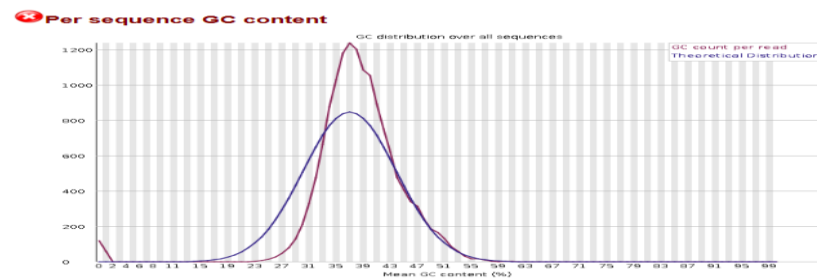
- Oui - c'est normal en séquençage
- Oui - qualité suffisante

Qualité par séquence



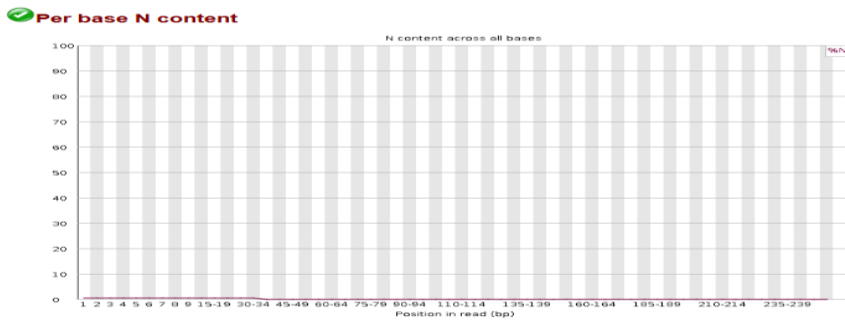
- Non - la majorité est de bonne qualité
- Score de 35-38 (excellente qualité)

Contenu en GC



- Oui - courbe en cloche symétrique
- Indique une contamination bactérienne ou des adaptateurs

Contenu en bases



- Oui - A≈T et G≈C
- Dû aux adaptateurs ou biais de PCR

Partie 4 :

Les données peuvent être utilisées pour l'analyse sans nettoyage préalable. La qualité est suffisante pour les analyses bioinformatiques standards (alignement, assemblage, variant calling).